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Exploring the complex immunomodulatory effects and gut defense via oral administration of *Astragali* radix water extract to normal mice

Mi-Gi Lee^{1†}, Youngju Song^{2†} and Hee Kang^{3*}

Abstract

Background *Astragali* radix (AR) is one of the most widely used traditional Chinese herbal medicines. It exhibits diverse biological activities, including immunomodulatory and anti-inflammatory properties; however, some of its activities have only been demonstrated in vitro.

Objective To examine the effects of orally administered AR extract on immune cells and the intestine under physiological conditions, which bridges the gap between previously observed in vitro outcomes and in vivo results.

Methods AR extract was prepared by hot water extraction. Three separate animal experiments were conducted to isolate macrophages, splenocytes, and the small intestine epithelium. For the macrophage preparation experiment, an intraperitoneal injection of sterile thioglycolate was administered. The mice received oral AR extract at doses of 0.1, 0.5, or 2.5 g/kg for ten days. At the end of each experiment, cells or tissues were isolated. A portion of macrophages and splenocytes were analyzed for the phenotypic changes. The remaining cells were cultured and stimulated with lipopolysaccharide (LPS) or mitogen ex vivo to assess activation status, proliferation, and cytokine production. Samples of the intestine were subjected to real-time RT-PCR.

Results Peritoneal macrophages from AR-treated mice exhibited increased expression of scavenger receptors, including SRA and CD36. Stimulation of these macrophages ex vivo with LPS selectively modulated the inflammatory response, including reduced expression of the costimulatory molecules CD40 and CD86, which are important for T cell responses, without affecting TNF- α and IL-6 production. Splenocytes from AR-treated mice exhibited a dose-dependent increase in CD4 and CD8 T cells; however, stimulation with mitogen decreased T cell proliferation and reduced IFN- γ production, which is essential for macrophage activation. An analysis of the small intestinal epithelium revealed an attenuated antimicrobial response, including reduced IgA content in the lumen and decreased expression of mucin-2 and polymeric Ig receptor genes.

Conclusion The response of immune cells following oral treatment with AR extract did not replicate the previously documented in vitro findings. Immune cells and intestinal epithelium from mice administered oral AR extract exhibited a selective anti-inflammatory phenotype. The overall findings indicate that the systemic effects after oral administration of AR extract include reduced sensitivity to inflammatory insults.

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Keywords Astragali, Macrophages, Splenocytes, Th1 response, Intestine, Antimicrobial

Introduction

Multicellular organisms are constantly exposed to microorganisms in their inner and outer environment. In response, immune cells have evolved with specialized receptors to efficiently detect these invaders. Immune cells are broadly categorized into innate and adaptive immune cells. Innate immune cells, including monocytes, macrophages, and dendritic cells, lack receptor specificity but can mount a rapid response. In contrast, adaptive immune cells, which consist of T cells (CD4 and CD8 T cells) and B cells, express highly specific receptors. The activation of T cells is contingent upon the activity of antigen-presenting cells, such as dendritic cells, macrophages, and B cells. Upon receiving appropriate signals from dendritic cells, CD4 T cells differentiate into distinct subsets, including Th1, Th2, Th17, and T reg cells. This categorization is dictated by the nature of the invading microbes, which influence the cytokine profiles produced by innate immune cells [1]. Th1 and Th2 cells, the earliest identified effector cells, may be characterized by interferon (IFN)- γ and interleukin (IL)-4 production, respectively. With respect to their effect on macrophages, the interplay between IFN- γ and IL-4 is reciprocal. IFN- γ contributes to the generation of inflammatory mediators by classically activating M1 macrophages, whereas IL-4 induces the polarization of M2 macrophages by mitigating this response [2].

The small intestine not only functions as a digestive tract but also participates in innate immunity. In direct contact with food and gut microbes, the epithelium is responsible for nutrient absorption and microbial defense [3]. An elegant study using the jejunum of B cell-deficient mice showed that the intestinal epithelium can upregulate the expression of many immune genes when B cells in the underlying connective tissue fail to produce an optimal immunoglobulin (Ig)A response [4]. These epithelial cells produce antimicrobial substances against commensal microbes and transport IgA to the lumen to block the entry of IgA-specific commensals [5].

Astragali radix (AR), the root of *Astragalus membranaceus* Bunge, is a commonly used traditional Chinese herbal medicine. Studies have demonstrated that AR and its ingredients exert beneficial effects on cardiovascular diseases, diabetes, respiratory diseases, and nervous system diseases and have antitumor, immunomodulatory, and anti-inflammatory properties [6]. AR contains over 200 identified constituents, including saponins, polysaccharides, and flavonoids [6]. The identification of its active components was done initially using in vitro models. This approach identified anti-inflammatory and immune-enhancing compounds in

AR [7–14]. Whether these active ingredients can reach cellular targets remains unclear because herbal medicines are administered orally. There is little evidence that herb-derived polysaccharides are absorbed into the systemic circulation in their native form. Most flavonoids and saponins exist as conjugates in the circulation. Pharmacokinetic studies of an AR water extract revealed a complex array of compounds and metabolites present in the circulation [15]. The intricate nature of this composition makes it challenging to attribute the effect of AR exclusively to any specific compound.

AR serves therapeutic purposes and is also considered a functional food, which appeals to individuals without specific ailments. Therefore, determining how AR affects normal immune cells following its oral administration is important. A study by Denzler et al. (2016) demonstrated acute changes in the number of circulating white blood cells, serum cytokines, body temperature, and blood pressure following the combined sublingual administration and ingestion of an ethanol-based AR extract to three healthy individuals; however, these changes returned to their baseline values within 24 h [16]. The study involved a single dose of AR and lacked an assessment of the functional immune cell response to inflammatory insults [16]. In our approach to bridging the gap between previously observed in vitro outcomes and in vivo contexts, we examined the response of macrophages and splenocytes isolated from normal mice that received an AR water extract for 10 days. The rationale for the 10-day treatment is consistent with the general practice of traditional herbal medicine practitioners, who commonly prescribe treatments lasting 10 to 15 days. We specifically selected a water extract because this solvent includes a polysaccharide fraction. In addition, we assessed the immune status of the small intestinal epithelium following oral administration. Our goal was to determine the effects of orally administered AR water extract on immune cells and the intestine under physiological conditions.

Materials and methods

Preparation of sample

AR was purchased from Damaon (Suwon, South Korea). Briefly, 200 g of the sample was ground and extracted with 2 L of deionized water in a reflux apparatus using a heating mantle for 2 h at 95 °C. After filtration, the extract was concentrated using a rotary evaporator and freeze-dried under vacuum. The yield of the AR extract was 21.9%.

High-performance liquid chromatography

For the standardization of the AR extract, formononetin was used as a quality control. Briefly, AR was dissolved in methanol (10 mg/mL) and the supernatant was obtained through centrifugation. Formononetin content was analyzed using a reverse-phase high-performance liquid chromatography (HPLC) system (Waters Alliance e2695, Milford, MA, USA) consisting of a photodiode array detector (2998, Waters) and equipped with a Capcell Pak C18 MG II column (5 μ m x 4.6 mm x 250 mm) (SHISEIDO, Osaka, Japan). Gradient flows for the two-solvent system (solvent A, 0.1% formic acid in water; solvent B, acetonitrile) were as follows: 90% A/10% B at 0 min, 85% A/15% B at 5 min, 50% A/50% B at 15 min, 50% A/50% B at 20 min, 40% A/60% B at 25 min, 40% A/60% B at 30 min, 15% A/85% B at 35 min, 15% A/85% B at 40 min, 10% A/90% B at 42 min, and 10% A/90% B at 45 min. The flow rate of the mobile phase was 0.8 mL/min with an injection volume of 10 μ L. Formononetin (Sigma, St. Louis, MO, USA) detection was done at 220 nm. The formononetin fraction from the AR extract was provisionally identified by spiking the input with authentic standards and comparing the retention times and UV–visible spectral patterns. The HPLC chromatograms are shown in Supplementary Figure S1. The formononetin quantity in the AR extract was determined to be 130 μ g/g extract.

Animals and experimental design

The animal protocol was approved by the Institutional Animal Care and Use Committee at Kyung Hee University [KHUASP(SE)-19–005]. Mice were cared for according to the specifications of the US National Research Council for the Care and Use of Laboratory Animals (1996). Seven-week-old male Balb/c mice were obtained from Koatech (Pyungtek, South Korea) and housed in a temperature- and humidity-controlled pathogen-free animal facility with a 12-h light–dark cycle. All animals underwent 1 week of adjustment before the experiments. Animals were randomly allocated to experimental groups, and there were no differences in body weight among the groups. Three distinct experiments were conducted to isolate macrophages (6 mice per group), splenocytes (5 mice per group), and small intestinal tissue (10 mice per group). For macrophage isolation, thioglycolate injection was administered as described below.

Dosage information

Doses for the animal experiments were based on previous studies [17, 18]. Mice were treated via oral gavage at doses of 0.1, 0.5, and 2.5 g/kg daily for 10 consecutive days. Translating these doses into human equivalents

corresponded to 0.05, 0.25, and 1.25 g/day for an average adult weighing 60 kg, based on a conversion factor between mouse and human [19]. Daily doses of raw AR in humans typically range from 1 to 25 g/day, which is equivalent to 0.21–5.25 g of the AR extract based on extract yield [16]. Thus, the doses used in the present study fall within the normal range. For oral gavage, the AR extract was prepared by dissolving in water at concentrations of 10, 50, and 250 mg/mL. At these concentrations, the volume per mouse was administered at 1 μ L/g of body weight. Control mice received an equal volume of water.

Macrophage preparation

To obtain a sufficient yield of macrophages from mice, we used thioglycolate. Mice were injected intraperitoneally with 2 mL of 3.5% sterile thioglycolate (BD, Sparks, MD, USA) 4 days before sacrifice. At the end of the experiment, mice were euthanized with carbon dioxide and peritoneal exudate cells were aseptically isolated by peritoneal lavage with cold DMEM (Hyclone, Logan, UT, USA) containing 10% fetal bovine serum (FBS; Hyclone) and 1% penicillin–streptomycin (WelGene, Kyungsan, South Korea). After centrifugation, cells were resuspended and counted using a TC20 Cell Counter (Bio-Rad Laboratories, Hercules, CA, USA). Peritoneal exudate cells were plated in 6-well plates and incubated overnight at 37°C. After the removal of non-adherent cells, attached cells were stimulated with 100 ng/mL lipopolysaccharide (LPS)(Sigma) for 24 h. Cells and supernatant were collected for subsequent assays.

Splenocyte preparation

Spleens were aseptically obtained for splenocyte isolation at the end of the experiment. After disrupting the spleen between glass slides in RPMI 1640 (Hyclone) with 1% FBS and 1% penicillin–streptomycin, the cells were filtered through a 70- μ m cell strainer. After centrifugation, red blood cells were lysed using BD PharmLyse lysing buffer (BD Biosciences, San Diego, CA, USA). Cells were resuspended in RPMI 1640 with 10% FBS and 1% penicillin–streptomycin and counted using a TC20 Cell Counter. Splenocytes were plated in 96-well plates (4 x 10⁵ per well) for the proliferation assay and 24-well plates (2 x 10⁶ per well) for cytokine analysis. After seeding, splenocytes were stimulated with 2 μ g/mL anti-CD3 antibody (BD Biosciences) for 48 h.

Flow cytometry analysis

Cells were washed twice in phosphate-buffered saline (PBS) and resuspended in FACS buffer (PBS/0.1% NaN₃/1% FBS). Cells were blocked with rat anti-mouse CD16/CD32 antibody (BD Biosciences) at 4°C for 5 min

and then stained with fluorescein-conjugated anti-mouse SRA (R&D Systems, Minneapolis, MN, USA), PE-conjugated anti-mouse CD36, FITC-conjugated CD11b, PE-conjugated anti-CD11b, PE-conjugated anti-mouse CD86, FITC-conjugated anti-mouse CD40, FITC-conjugated anti-mouse CD3, PE-conjugated anti-mouse B220, FITC-conjugated anti-mouse CD4, PE-conjugated anti-mouse CD8a (BD Biosciences) (all antibodies were diluted 1:100) for 30 min on ice in the dark. Matched isotype antibodies were used to show non-specific binding. The cells were washed and resuspended in FACS buffer. 10,000 events were acquired on a Navios flow cytometer (Beckman Coulter, La Brea, CA, USA), and the data were processed using Kaluza software (Beckman Coulter).

Proliferation assay

The proliferation of splenocytes was measured using the MTS assay. Briefly, splenocytes were plated in 96-well plates and stimulated with 2 µg/mL anti-CD3 antibody (BD Biosciences) for 48 h. Then, 20 µL of the CellTiter96® Aqueous One Solution (Promega, Madison, WI, USA) was added to the well, and the quantity of formazan product was measured using an iMark microplate reader (Bio-Rad Laboratories, Hercules, CA, USA) at 490 nm.

Isolation of Epithelial tissue and sampling of intestinal content

After the mice were sacrificed by cervical dislocation, the small intestines were harvested, and the mesentery, fat tissue, and Peyer's patches were removed. Epithelial cells were isolated as described [20] with some modifications. In brief, one gram of the intestine was obtained and cleared of its contents by flushing with a syringe filled with cold PBS. This intestinal content was centrifuged at 10,000xg for 20 min at 4 °C and stored at -20 °C for further analysis. The remaining intestine was opened longitudinally, cut into 4–5-cm lengths, and incubated in Hank's Balanced Salt Solution (HBSS) (Sigma) with 2% FBS and 5 mM DTT (Sigma) for 20 min at 37 °C with continuous stirring. The solution was removed using a 100-µm cell strainer, and the remaining tissue was incubated in HBSS with 2% FBS and 5 mM EDTA for 15 min at 37 °C. Cells were collected using the cell strainer. This step was repeated three times. The cells were then pooled and used for RNA extraction.

Enzyme-linked immunosorbent assay (ELISA)

The cytokine levels in supernatants were determined using BD OptEIA mouse ELISA sets (BD Biosciences) according to the manufacturer's protocol. For IgA quantification, capture anti-mouse IgA antibody, detection anti-mouse IgA antibody, streptavidin–horseradish peroxidase, and standard mouse IgA were purchased from

BD Biosciences. ELISA plates were coated with capture antibody overnight at 4 °C, washed, and blocked with 10% FBS in PBS for 1 h at room temperature. After washing, diluted samples and standard IgA were added to each well for 2 h. After washing, the detection antibody was added to the well for 1 h, followed by a 30 min-incubation with streptavidin–horseradish peroxidase. TMB (BD Biosciences) was added after washing. The stop solution was added after 30 min. Absorbance at 450–570 nm was determined using an iMark microplate reader.

RNA isolation and real-time PCR

RNA was extracted using an RNeasy mini kit (Qiagen, Germany) and reverse-transcribed into cDNA using the High Capacity cDNA Reverse Transcription kit (Applied Biosystems, Foster, CA, USA). Real-time PCR was performed using SYBR Green Mix (Applied Biosystems) on a StepPlus One real-time PCR system (Applied Biosystems). Quantifying each cDNA copy number was based on the standard curve calculation method. Target genes were normalized to a housekeeping gene glyceraldehyde-3-phosphate dehydrogenase (GAPDH). Primer sequences are shown in Supplementary Table S1.

Statistical analysis

All statistical analyses were performed with IBM SPSS 28.0 version software (IBM, Chicago, IL, USA). Data are presented as the mean ± SD. The differences between groups were compared using a Student's t-test or analysis of variance. If the statistical analysis showed that differences between multiple groups were significant, the Dunnett or LSD test was used for further comparison. *P*-values of <0.05 were considered statistically significant.

Results

Enhanced scavenger receptor expression in peritoneal macrophages following the oral administration of AR extract post-thioglycolate injection

Peritoneal macrophages, obtained via intraperitoneal injection of thioglycolate, differentiate from blood monocytes and express scavenger receptors such as CD36 and SRA on their cell surface [21, 22]. Increased expression of scavenger receptors is characteristic of M2 macrophages, which have a wound-healing or anti-inflammatory phenotype that contributes to inflammation resolution and apoptotic cell clearance [23]. Before assessing the functional response of peritoneal macrophages, we examined the effect of an orally administered AR extract on scavenger receptor expression. CD11b serves as a macrophage marker. Using flow cytometry, we double-stained cells for the expression of CD36(+)CD11b(+) or SRA(+)CD11b(+), which represent monocyte-derived macrophages.

Most peritoneal macrophages in the control group expressed CD36 (96.6%), indicating an M2 phenotype, which may be involved in dead cell clearance following thioglycolate injection (Fig. 1A, C). Treatment with the AR extract tended to decrease this population (96.1%, 95.3%, and 94.5% in 0.1, 0.5, and 2.5 g/kg groups, respectively), with a significant reduction in

the high-dose group. This decrease in the percentage of CD36(+)CD11b(+) cells may have resulted from a tempered response to the intraperitoneal injection of thioglycolate. However, the mean fluorescence intensity (MFI) of CD36, an indicator of protein expression on the cell surface, did not decrease in the AR groups, whereas the low-dose group exhibited higher

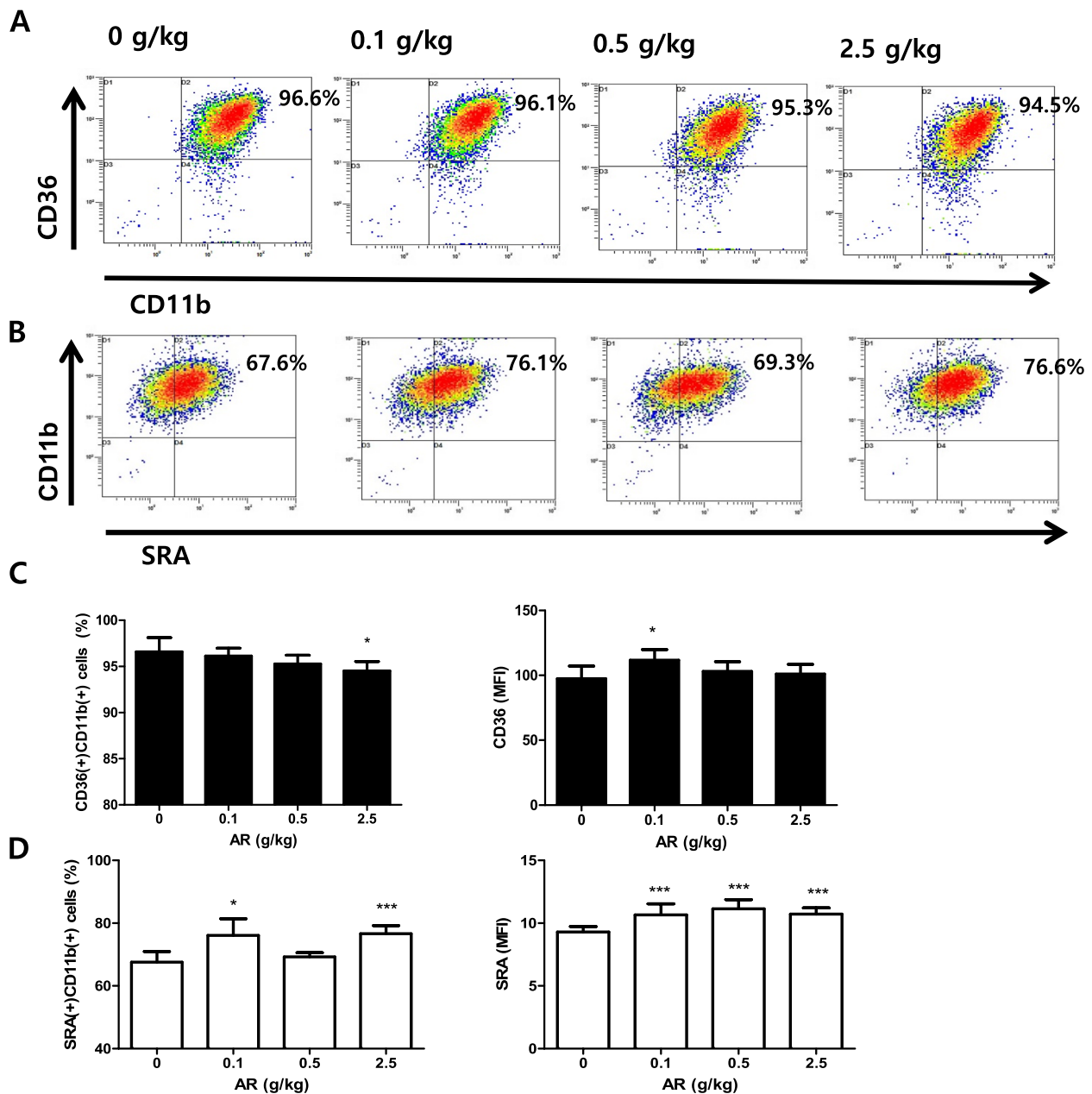


Fig. 1 Enhanced scavenger receptor expression after the oral administration of *Astragali radix* extract (AR). Mice were orally administered AR (0.1, 0.5, or 2 g/kg) for 10 days. **A, B** Peritoneal macrophages were isolated from thioglycolate-injected mice and double-stained with FITC-conjugated anti-CD11b antibody (ab) and PE-conjugated anti-CD36 ab or FITC-conjugated anti-SRA ab and PE-conjugated anti-CD11b ab. Representative dot plots are shown. **C, D** Bars represent the percentage of the double-stained cell populations and the mean fluorescent intensity (MFI) of CD36 and SRA (n=6 per group). * $P < 0.05$, *** $P < 0.005$ versus Control (0 g/kg)

values. Thus, despite a reduction in the percentage of CD36(+)CD11b(+) cells, CD36 expression per cell increased somewhat. SRA expression was less predominant compared with CD36, with 67.6% SRA(+)CD11b(+) cells in the control group (Fig. 1B, D). Treatment with the AR extract increased this cell population (76.1%, 69.3%, and 76.6% at 0.1, 0.5, and 2.5 g/kg doses, respectively). SRA MFI also increased at all AR doses, indicating enhanced SRA expression per cell. Overall, AR treatment appears to confer an anti-inflammatory phenotype during thioglycolate-induced peritonitis while enhancing the expression of scavenger receptors by monocyte-derived macrophages.

Oral administration of AR extract has no effect on the production of inflammatory cytokines by macrophages

Previous studies have demonstrated that in vitro treatment of AR fractions reduced the production of TNF- α and IL-6 by LPS-stimulated macrophages [7–9, 12, 14]. To determine whether a similar observation could be made in macrophages isolated from AR-treated mice, we cultured these macrophages and stimulated them with LPS. Despite testing various doses, no significant reduction in TNF- α secretion was observed compared with the control group (Fig. 2). Notably, IL-6 levels decreased only in the 500 mg/kg group. Conversely, the high-dose group showed an increase. Our data from macrophages isolated from mice orally administered the AR extract did not replicate the inhibition of inflammatory cytokines observed in previous in vitro studies [7–9].

Oral administration of AR extract suppresses the expression of LPS-induced costimulatory molecules in macrophages

Costimulatory molecules, such as CD86 and CD40 that are present on the cell surface of macrophages, deliver activating signals to T cells through their interaction with CD28 and CD40L, respectively, on T cells. We analyzed whether an oral AR water extract affected the expression of these costimulatory molecules induced by LPS ex vivo. Macrophages from the control group exhibited an increase in CD86 and CD40 expression following LPS stimulation as indicated by the MFI (Fig. 3). Conversely, in the AR group, the expression of these costimulatory molecules decreased in a dose-dependent manner. These findings show that oral AR extract selectively alters the activation status of macrophages.

AR extract has the potential to inhibit the Th1 response

Before evaluating the splenic T cell functional response, we determined the effect of orally administered AR extract on the population of splenic T and B cells. Flow cytometric analysis revealed a dose-dependent increase in the percentage of T cells, concomitant with a dose-dependent decrease in the percentage of B cells (Fig. 4A, C). Further examination of T cell subsets revealed that CD4 T cells and CD8 T cells increased in a dose-dependent manner (Fig. 4B, D). Subsequently, an equal number of splenocytes from the control and AR groups were stimulated with T cell polyclonal mitogen anti-CD3 antibody for 48 h, and proliferation and cytokine production were evaluated. Splenocytes from the AR group exhibited a dose-dependent decrease in proliferation (Fig. 5A). An interesting pattern emerged

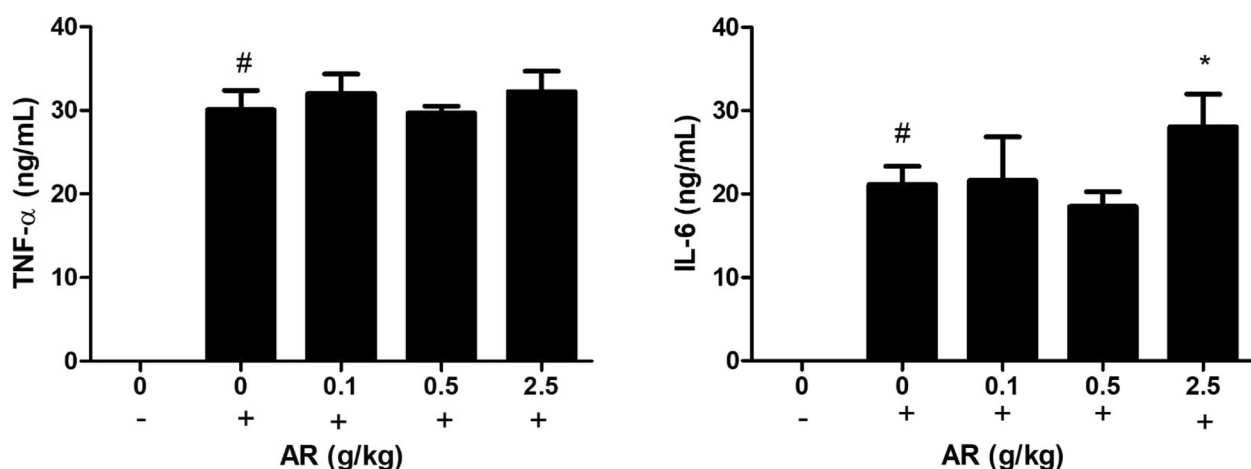


Fig. 2 The oral administration of AR did not suppress the production of LPS-induced inflammatory cytokines in macrophages. Macrophages isolated from the control and AR groups were stimulated with LPS (100 ng/mL) for 24 h. Tumor necrosis factor (TNF)- α and interleukin (IL)-6 levels in the supernatant were measured by ELISA ($n=6$ per group). [#] $P<0.005$ versus Control (0 g/kg, -), ^{*} $P<0.05$ versus Control (0 g/kg, +)

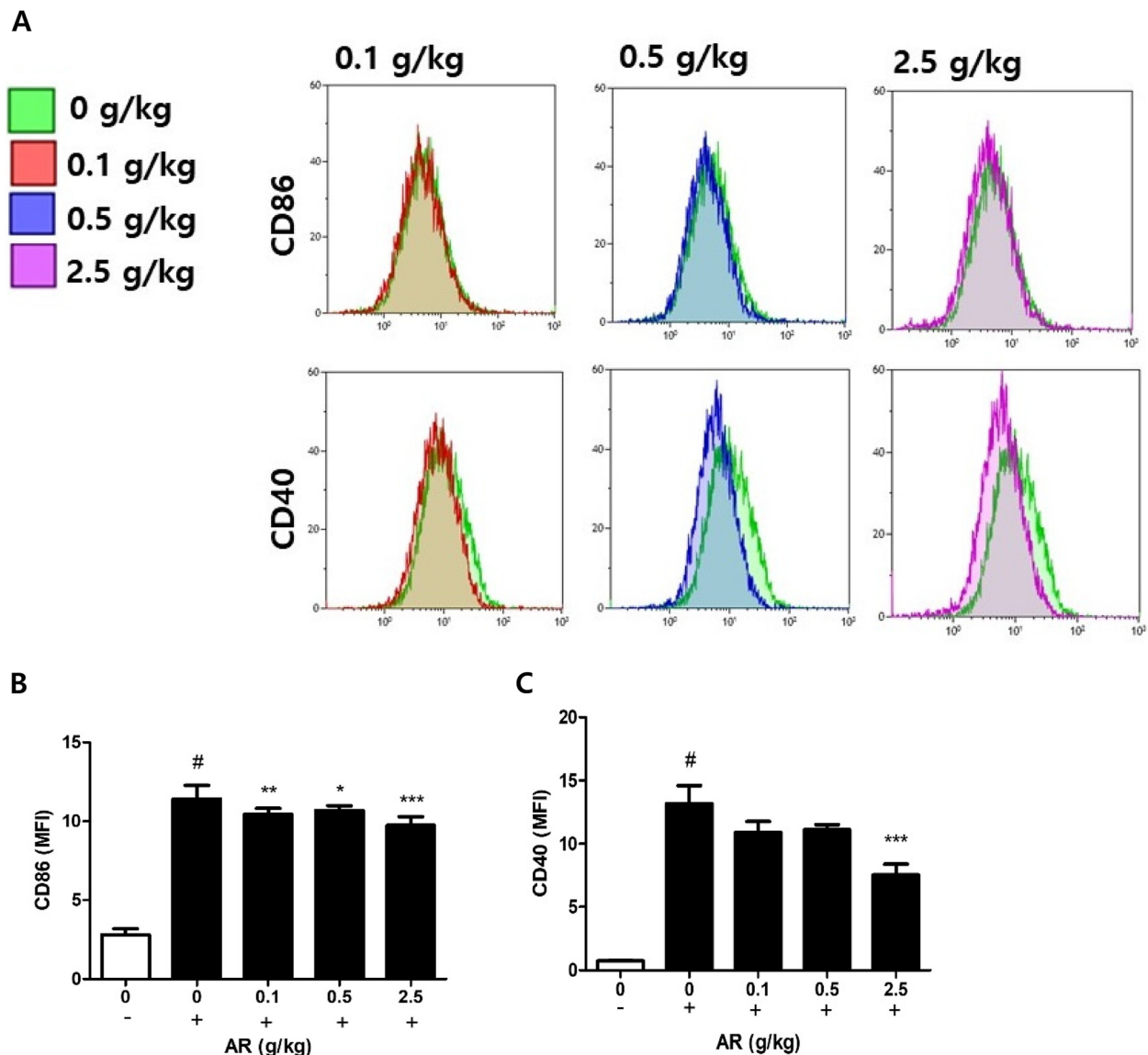


Fig. 3 The oral administration of AR decreased the surface expression of LPS-induced costimulatory molecules. Macrophages isolated from the control and AR groups were stimulated with 100 ng/mL LPS for 24 h and stained with PE-conjugated anti-CD86 ab or FITC-conjugated anti-CD40 ab. **A, B** Representative overlaid histograms are shown. **C** Bars represent the percentage of the MFI of CD86 and CD40 ($n=6$ per group). (-): without LPS, (+): LPS treatment. # $P<0.005$ versus Control (-), * $P<0.05$, ** $P<0.01$, *** $P<0.005$ versus Control (+)

in the cytokine levels produced by T cell stimulation. Notably, in the lowest dose group, IL-2 production decreased; however, with increasing doses, it gradually rebounded to levels comparable with those of the control group (Fig. 5B). Conversely, IFN- γ production exhibited a consistent decrease at all tested doses (Fig. 5C). In contrast, IL-4 showed a tendency to increase, albeit without reaching statistical significance (Fig. 5D). Overall, oral AR extract showed the potential to inhibit the Th1 response.

Decreases in intestinal IgA and antimicrobial gene expression from AR-treated mice

We determined whether the oral administration of AR affects the immune status of the small intestinal epithelium. The measurement of IgA content in the intestinal luminal wash revealed a significant decrease in the AR group compared with the control group (Fig. 6A). The analysis of the polymeric Ig receptor (pIgR) gene expression, a transporter that delivers IgA from the lamina propria to the upper layer of the epithelium, revealed a

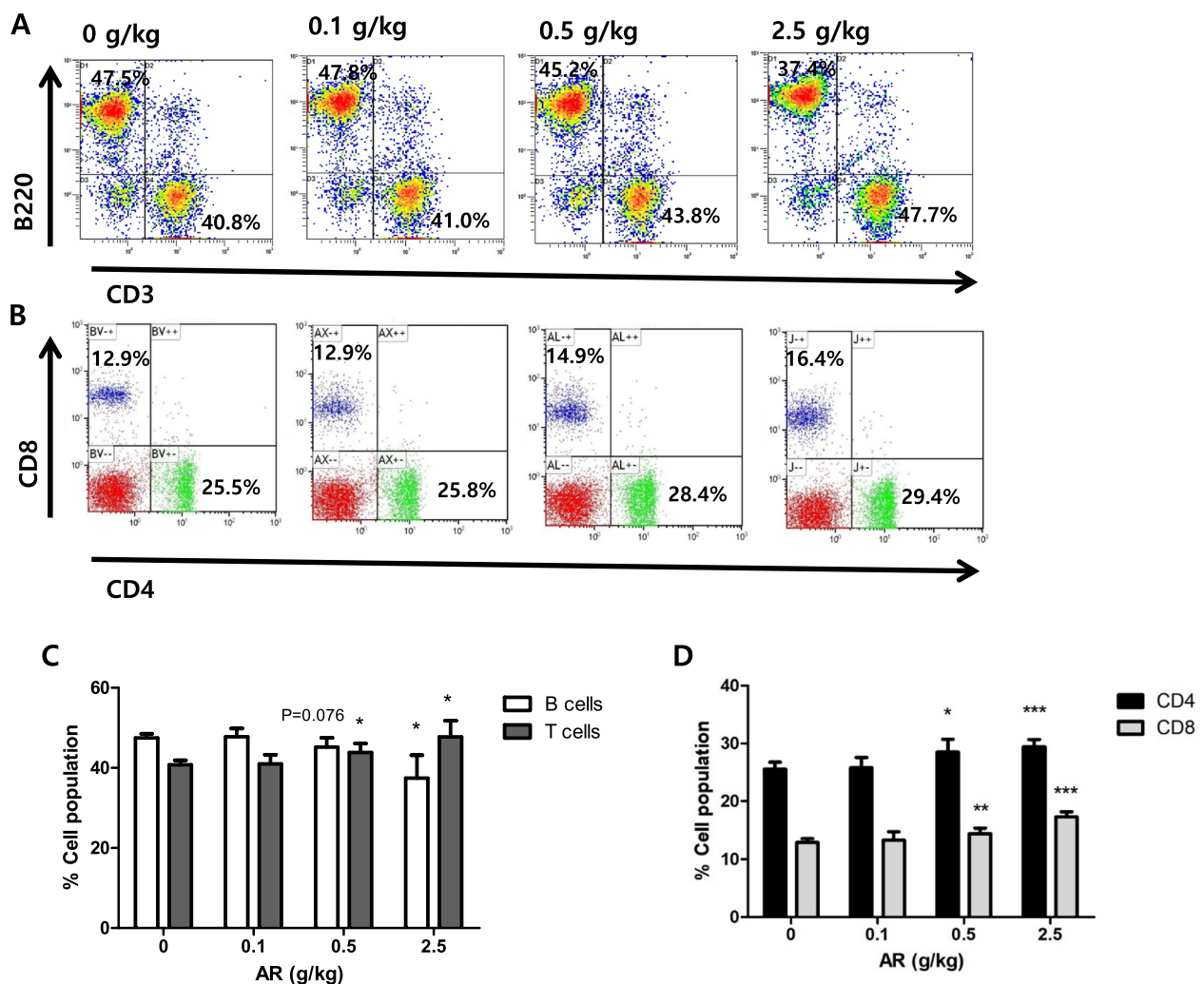


Fig. 4 Dose-dependent increase in splenic T cells after the oral administration of AR. Splenocytes were isolated from the control and AR groups and double-stained with FITC-conjugated anti-CD3 antibody and PE-conjugated anti-B220 antibody for T cell and B cell markers, respectively, or FITC-conjugated anti-CD4 antibody and PE-conjugated anti-CD8 antibody. **A, B** Representative dot plots are shown. **C, D** Bars represent the percentage of CD4(+) cells or CD8(+) cells ($n=5$ per group). * $P<0.05$, $P<0.01$, *** $P<0.005$ versus Control

decrease in the AR group ($P=0.051$) [24]. The expression of the mucin 2 gene, which forms a physical barrier in the epithelium, also significantly decreased in the AR group ($P=0.0007$) [25]. The antimicrobial gene Reg3y, which targets Gram-positive bacteria, did not show a significant change, and there was a tendency toward a decrease ($P=0.142$) [25]. The results indicate a reduced antimicrobial response in the small intestinal epithelium following the oral administration of the AR extract.

Discussion

By isolating immune cells from mice that received an oral AR extract, this study revealed that macrophages and splenocytes do not exhibit the same biological

activities as previously documented in vitro. In addition, we observed an effect on the immune status of the small intestinal epithelium, highlighting the broad spectrum of immunomodulatory effects exerted by the AR extract.

In this study, a significant finding emerged regarding the potential of AR extract to enhance scavenger receptor expression in peritoneal macrophages obtained by thioglycolate-induced peritonitis. The thioglycolate medium is well-known for inducing sterile inflammation by recruiting blood monocytes to the peritoneum, where they differentiate into tissue macrophages responsible for clearing neutrophils that arrive earlier in the peritoneal cavity [26, 27]. Consequently, peritoneal macrophages 3 to 4 days post-thioglycolate

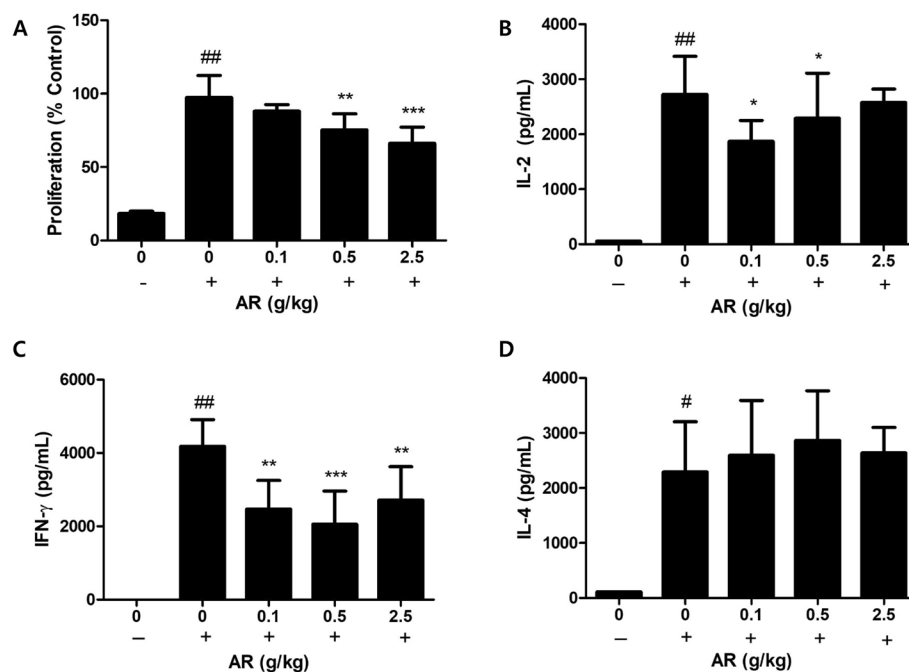


Fig. 5 Alterations in anti-CD3 ab-induced proliferation and cytokine production in splenocytes after the oral administration of AR. Splenocytes isolated from control and AR groups were stimulated with anti-CD3 antibody (2 µg/mL) for 48 h ($n = 5$ per group). **A** The proliferation of the splenocytes was measured using the MTS assay. **B–D** Cytokine production at 48 h of stimulation was measured by ELISA. (–): without anti-CD3 ab, (+): anti-CD3 ab treatment. # $P < 0.01$, ## $P < 0.001$ versus Control (–), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.005$ versus Control (+)

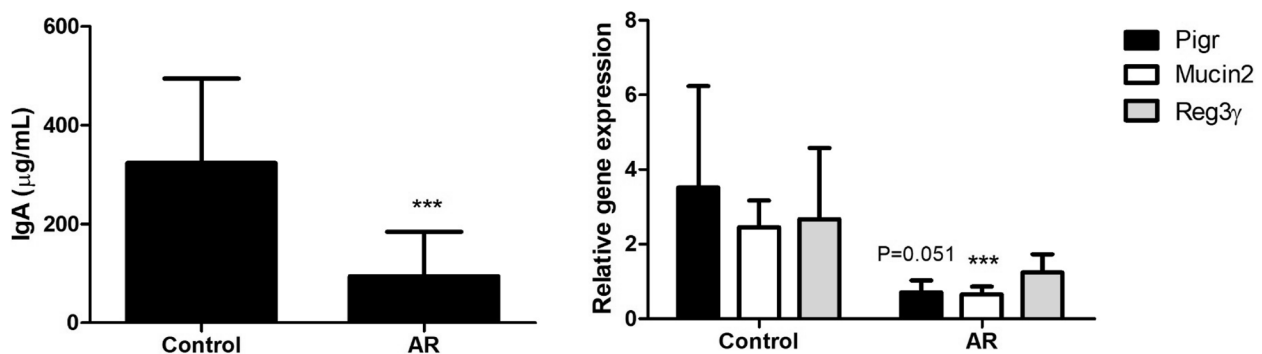


Fig. 6 Reduced IgA and antimicrobial gene expression in the small intestine of mice treated with AR. **A** Luminal fluid was obtained from the small intestine, and IgA was analyzed by ELISA ($n = 10$ per group). **B** Epithelial cells were isolated from the small intestine. RNA was extracted, and qPCR was performed. Target genes were normalized to GAPDH levels ($n = 5$ per group). *** $P < 0.005$ versus Control

injection may be categorized as belonging to the M2 phenotype. Our observation revealed that while most peritoneal macrophages express CD36 on their cell surface, only two-thirds express SRA. Notably, CD36(+) CD11b(+) cells likely represent the entire population of monocyte-derived macrophages. The reduced percentage of CD36(+)CD11b(+) cells in the AR-treated groups may result from a tempered response to thioglycolate-induced peritonitis; however, the observed increase in the MFI of CD36 and SRA and the

percentage of SRA(+)CD11b(+) cells suggests that the oral administration of AR extract may contribute to the recovery from thioglycolate-induced peritonitis.

LPS is recognized by TLR4, which subsequently activates NF-κB and IRF-3 through the recruitment of the adaptor molecules, MyD88 or TRIF [1]. This results in the expression of molecules essential for inflammatory and immune responses. Notably, the signaling pathway leading to the upregulation of costimulatory molecules differs from that of TNF-α and IL-6 [28]. CD40 and CD86

expression relies on the TRIF-IRF3-type I IFN axis [28]. In the present study, macrophages from AR-treated mice exhibited diminished surface levels of CD40 and CD86, with no apparent effect on TNF- α and IL-6 production. This selective modulation suggests that bioavailable AR compounds or their metabolites may enter macrophages and act downstream of IRF. However, the absence of a reduction in TNF- α and IL-6 contradicts previous findings in macrophages exposed to the AR extract or its compounds, which exhibited an overall decrease [7–9, 12–14]. These divergent outcomes highlight the gap between in vitro and in vivo treatments. Additionally, in various animal models administered oral AR, downregulation of TNF- α and IL-6 was observed at the tissue level [17, 29, 30]. Environmental factors, such as debris from damaged cells in wounded tissue can trigger the expression of inflammatory cytokines. Considering the potential of AR to promote the scavenging M2 phenotype (Fig. 1), the inhibition of inflammatory cytokine expression in disease models following AR administration may be attributed to macrophages with an enhanced capacity to clear cellular debris within the affected lesion.

The spleen is recognized as a reservoir that can mobilize immune cells into the circulation and subsequently toward peripheral tissues as necessary [31]. The observed increase in the percentage of CD4 and CD8 T cells and the decrease in B cells within the spleen of mice treated with AR extract may be associated with distinct mobilization kinetics. In a study by Denzler et al., a single administration of AR resulted in the rapid mobilization of immune cells, with the peak of each cell population varying throughout the day [16]. Thus, the AR extract may stimulate the mobilization of B cells and other non-T cells more rapidly, resulting in a higher proportion of T cells remaining in the spleen at the time of isolation. This possibility merits further study.

Despite a dose-dependent increase in splenic CD4 and CD8 T cells, the proliferation of splenocytes induced by anti-CD3 antibody exhibited a dose-dependent decrease. This contrasts with previously reported observations from studies on the effects of AR saponins, which showed an increase in splenic T cell proliferation [32–34]. The higher doses of AR extract may contain compounds that inhibit T cell proliferation. Alternatively, we hypothesize that the dose-dependent decrease in T cell proliferation may be attributed to a reduced number of accessory cells, which are splenic adherent cell fractions, such as macrophages and dendritic cells. For the mitogen activation of T cells in vitro, the interaction between T cells and accessory cells is necessary [35]. Accessory cells provide costimulatory signals, such as CD80 and CD86,

by making direct contact with T cells [36]. Because splenocytes from the AR-treated group contained a higher proportion of T cells as doses increased, it is plausible that fewer accessory cells were available to provide costimulatory signals, thereby explaining the dose-dependent proliferation. Interestingly, the pattern observed in IL-2 production was opposite to that of the proliferation response. This suggests that orally administered AR extract distinctly affects the pathway governing proliferation and IL-2 production.

According to the study by Denzler et al., the acute administration of AR to humans appeared to increase the production of inflammatory cytokines, such as IL-6, IL-13, IFN- γ , and TNF- α , in their serum, which was associated with the Th1 response, as suggested by the authors [16]. However, detecting IL-4 in serum poses a challenge because of its rapid T cell consumption. In the present study, splenocytes from AR-treated mice exhibited a diminished Th1 response, as evidenced by a decrease in IFN- γ production. These results are consistent with an earlier study demonstrating that AR administration to mice decreased serum levels of IFN- γ following systemic stimulation of T cells and accompanied by an increase in IL-4 [37]. Notably, reduced IFN- γ production by activated splenic T cells occurred in a dose independent manner. It appears that the absorption of unidentified active compounds capable of inhibiting CD3-induced IFN- γ production in T cells reaches saturation. The diverse patterns observed in proliferation, IL-2, and IFN- γ suggest that multiple active compounds within the AR extract exert different effects on T cell activation signaling. Moreover, the oral administration of AR extract selectively inhibits crosstalk between macrophages and T cells. This is evidenced by the suppression of costimulatory molecules expressed by macrophages and the reduction of IFN- γ production by T cells.

Epithelial cells within the small intestine express antimicrobial genes through the TLR/MyD88 pathway and respond to commensal and pathogenic microorganisms [25, 38]. In the small intestinal epithelial tissues of AR-treated mice, we observed indications of a decreased microbial burden, as evidenced by reductions in luminal IgA content and the downregulation of pIgR and mucin2. The results suggest that the AR extract may alter the gut microbial environment, potentially dampening the epithelial TLR/MyD88 pathway. Improved gut mobility and digestion may influence microbial composition. This reduction in defensive burden on the intestinal epithelial cells may enable a greater allocation of energy toward metabolic function [4]. Because qi-tonifying herbs like AR are believed to boost immunity and affect overall well-being,

further studies are warranted to fully elucidate the complete spectrum of activities exhibited by AR extract.

Conclusion

This study showed that the response of immune cells following oral treatment with AR extract did not replicate the previously documented findings *in vitro*. Immune cells from mice receiving oral AR water extract exhibited a selective anti-inflammatory phenotype. In addition, the AR extract decreased the expression of antimicrobial genes in the intestinal epithelium. Combining all these results, the systemic effects after oral administration of AR extract include reduced sensitivity to inflammatory insults. A limitation of this study was the absence of a confirmed causal relationship between systemic immunity and intestinal defense. Further studies are needed to bridge these gaps and provide a more comprehensive understanding of the potential benefits of AR.

Abbreviations

AR	Astragali Radix
FBS	Fetal bovine serum
ELISA	Enzyme-linked immunosorbent assay
IFN	Interferon
Ig	Immunoglobulin
HPLC	Hyper-performance liquid chromatography
IL	Interleukin
LPS	Lipopolysaccharide
MFI	Mean fluorescence intensity
PBS	Phosphate buffered saline
plgR	Polymeric immunoglobulin receptor
Th	T helper
TNF- α	Tumor necrosis factor- α

Supplementary Information

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Supplementary Material 1.

Authors' contributions

ML and YS performed the study and wrote the original manuscript. HK designed the study and reviewed the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials

The data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

The animal protocol was approved by the Institutional Animal Care and Use Committee at Kyung Hee University (KHUASP(SE)-19-005), and mice were cared for according to US National Research Council for the Care and Use of Laboratory Animals (1996) specifications.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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